

High and Low Achieving Education Students on Processing, Retaining, and Retrieval of Information

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High Achievers (students whose Educational Psychology grades were 90% and above) and Low Achievers (students whose grades were 79% and below) were compared on how they process, retrain, and retrieve information. The Information of Learning Processes, an instrument used to collect the data, consists of four independent scales: Deep Processing, Elaborative Processing, Fact Retention, and Methodical Study, showed differences between the two groups on Deep Processing and Fact Retention scales. In each case, the High Achievers reported significantly higher scores than the Low Achievers. The data can be interpreted that the High Achievers analyze information, retain and retrieve it better than do the Low Achievers. The question raised is how will these learning styles affect these students (who are potential teachers) in teaching and evaluation of their students.

Educational psychologists and researchers have attempted to understand how students differed in processing, retaining, and retrieval of the information. To explore these differences, researchers used various types of personality, attitudinal, cognitive style, and ability measures (Cowell & Entwistle, 1971; Cropley & Field, 1969; Schmeck, 1983). Most researchers agreed that learning is related to one's personality, attitude, and thinking. They also agreed that learning strategies are modifiable due to one's perception of how information learned is to be measured and evaluated. However, some of these earlier learning measures were not very useful in assessing classroom activities (Schmeck, 1983).

Consistent with the thinking of Craik and Lockhart (1972) that memory is a 'by

product of thinking: traces left behind by past information-processing,' Schmeck (1983) defined learning strategy as "a pattern of how information-processing activities are used to prepare for an anticipated test of memory" (p.234). Schmeck (1983) agreed with Tallmadge and Shearer (1969, 1971) that the "learning style would be a more useful concept than the traditional personality and cognitive style constructs in accounting for the variances in academic performances" (p.233).

Agreeing with Craik and Lockhart (1972) on their levels of processing model, Pask's (1976) operational learning strategies, and achievement motivation theory, Schmeck, Ribich, and Ramanaiah, (1977) saw the need for developing a learning assessment from a behavioral-process orientation. They developed the Inventory of Learning Processes (ILP) which assesses information-processes in academic settings. That is, the inventory assesses how students' process, retain, and retrieve the information they study.

The ILP provides four independent scale scores (Schmeck et al., 1977). The *Deep Processing* (DP) scale assesses the extent to which one critically evaluates, conceptually organizes, and compare and contrasts infor-

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mation under study. The *Elaborative Processing* (EP) scale assesses the extent to which one translates new information into his/her own terminology. The *Fact Retention* (FR) scale assesses how one processes specific factual information. The *Methodical Study* (MS) scale assesses whether one uses systematic techniques recommended in 'how-to-study' manuals.

This inventory has been used extensively throughout the country in various classroom settings. Numerous studies (Albaili, 1993; Gadzella, 1995; Gadzella, Ginther, & Williamson, 1986; Gadzella, Stephens, & Baloglu, 2002; Miller, Alway, & McKinney, 1987; Schmeck, 1982; Schmeck & Grove, 1979; Schmeck & Phillips, 1982; Schmeck et al., 1997) have shown that there are significant relationships between students' learning style responses and their course grades and GPAs, respectively. However, are there differences between high and low academic achievers on the ILP scores.

In one study (Gadzella, 1995), scores on the ILP scales (for 86 freshmen enrolled in psychology classes) were compared with the students' course grades. Data showed that students who earned A grades in the course (compared to students who earned B, C, or D grades) reported significantly higher scores on the DP, EP, and MS scales of the ILP. In another study (Gadzella et al., 1987), the median of the students' GPA was used to identify High and Low Academic Achievers (for 158 students enrolled in Psychology classes). The responses to the ILP scales were used to compare differences between the High and Low Academic Achievers. The results showed that the High Achievers reported significantly higher scores on DP and FR scales. A study (Schmeck & Grove, 1979) on relationships (for 790 college students) between GPAs and ILP scores showed that students with high GPAs reported high scores on DP, EP, and FR scales. Similar results were found in another study

(Schmeck, 1983) in that, high academic achievers tended to score high on DP, EP, and FR scales of the ILP.

In the above mentioned studies, subjects were pursuing higher education in colleges and universities but no mention was made as to their majors or possible careers. The focus of the present study was on students who were potential teachers. How do they process the information that they study?

Teacher training institutions require students to take several courses to prepare them for the teaching profession. One such course is Educational Psychology. Usually, the course includes subject areas such as: research and methodology, moral and cognitive development, learning theories, and measurement and evaluation.

In a recent study (Gadzella et al., 2002) with 105 students enrolled in Educational Psychology classes, data showed significant relationships between two ILP scale scores (DP and FR) and Educational Psychology course grades. The average course grade was a B. In the present study, the purpose was to investigate whether there were differences between students who earned A grades (High Achievers) in Educational Psychology and students who earned C or D grades (Low Achievers) in the course on the ILP scales. Specifically, the aim was to determine if there were differences between students, identified as High and Low Achievers in Educational Psychology, on how they process, retain, and retrieve the information they study.

Method

Subjects. There were 61 students majoring in Education, of which 38 (one man and 37 women) were referred to as High Achievers and 23 students (7 men and 16 women) were referred to as Low Achievers. Their ages ranged from 19 to 54 years ($M = 28.6$, $SD = 8.4$). In this group, there were 10 sophomores, 33 juniors, 8 seniors, and 9 graduates. One person did not report the college classification.

Instrument. The Inventory of Learning Processes, ILP, (Schmeck et al., 1977) was used to collect the data. The instrument is a self-reporting questionnaire with 62-items, assessing one's style of processing information. The four independent scales derived from the ILP were described above. The reliability and validity of the ILP scales have been studied and reported in detail. For instance, Schmeck et al. (1977) reported internal consistencies for the four ILP scales ranging from .52 to .82 and test retest reliabilities ranging from .78 to .88. Albaili (1993) reported test-retest reliability coefficients for ILP scales ranging from .68 to .80 and House and Gadzella (1995) reported test-retest reliabilities ranging from .79 to .88.

In 1977, Schmeck et al. reported significant correlations between multi-choice psychology test and scores on the DP and EP scales ($r = .42$ and $r = .51$, respectively). In another study (Schmeck et al., 1977), data showed significant relationships between memory on a word-list test and scores on DP ($r = .59$) and EP ($r = .35$) scales. Bartling

(1988) reported predictive validity coefficients of the ILP scales by correlating them with college and high school GPAs and ACT scores. The correlations ranged from .34 to .58. A number of studies (Gadzella, Ginther, & Williamson, 1987; Miller et al., 1987; Schmeck & Grove, 1979) have shown good predictive validity coefficients for the individual ILP scales with reference to the GPAs.

Procedure. Students responded to the ILP during class periods. They signed a research release form indicating that their course grades and responses to the ILP may be used for research purposes. They received bonus points for participating in the study. The average of the four tests administered in Educational Psychology classes was the student's course grade. Students whose average grades on the tests were 90% and above received A grades and were referred to as High Achievers. Students who received an average grade of 79% and below received C/D grades and were referred to as Low Achievers. For the two groups, *t*-tests were used to analyze the responses to each of the four ILP scales.

Table 1
Means, Standard Deviations, and t-tests on the Inventory of Learning Style Scales for High and Low Achievers (df = 2/59)

Inventory of Learning Style Scale	Achiever Group	N	M	SD	t-test
Deep Processing	High	38	12.11	3.76	2.23*
	Low	23	9.70	4.32	
Elaborative Processing	High	38	10.21	2.63	1.19
	Low	23	9.35	2.92	
Fact Retention	High	38	5.76	1.15	3.19**
	Low	23	.74	1.32	
Methodical Study	High	38	10.16	4.56	1.18
	Low	23	8.70	4.89	

* $p < .03$ ** $p < .01$

Results

Means, standard deviation, and t-tests for the two groups on the four ILP scales are presented in Table 1. The data show that the High Achievers reported significantly higher scores than the Low Achievers on two ILP scales: Deep Processing and Fact Retention. What does this mean?

Discussion and Conclusion

The tests administered in the Educational Psychology classes consisted of objective-type items which measured primarily two types of learning processes: deep processing (analyzing information) and retention of facts. The data from the Deep Processing scale mean that the High Achievers (more than the Low Achievers) evaluate the information that they study more critically, organize it conceptually, and make comparisons and contrasts. The data from the Fact Retention scale indicate that the High Achievers process, retain, and retrieve specific information (such as, dates of special events, etc.) more effectively than do the Low Achievers. These findings concur with those previously cited (Gadzella, 1995; Gadzella et al., 1987; Schmeck & Grove, 1979; Schmeck, 1983) that High Achievers report higher scores on Deep Processing and Fact Retention scales. From the findings in the present study, one can conclude that High and Low Achievers in Educational Psychology (both potential teachers) process these types of information differently. The question that can be raised is, how will it reflect their teaching and evaluation of their students on these types of learning and evaluation.

Further studies should be encouraged to determine how essay-type tests and activities, such as projects on computers or laboratory assignments are evaluated for High and Low Achievers in Educational Psychology classes. Some students prefer to use their own words to indicate what they know. Therefore, they might perform better on

essay-type tests and/or on projects than they do on objective-type tests. The two scale scores, Elaborative Processing and Methodical Study of the ILP, might correlate more effectively with measures on the essay-type tests and/or projects. Lockhart and Schmeck (1983) showed that there are significant relationships between the different types of classroom measures and evaluations with the Elaborative Processing and Methodical Study scales as well as with the Deep Processing and Fact Retention scales. In addition, future studies should be conducted with larger number of students including those aiming to teach different subject matter and at different levels.

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